

UNIVERSITY OF **WATERLOO**



Department of Mechanical and Mechatronics Engineering

DESIGN OF TEST STAND FOR HIGH RISK ROCKET ENGINE TESTING

A Report Prepared For:

The University of Waterloo

Prepared By:

Hamza Abuabah

200 University Ave W.

Waterloo, Ontario, N2L 3G1

May 19th, 2020

Hamza Abuabah
200 University Ave W.
Waterloo, Ontario, N2L 3G1
May 19th, 2020

Prof Michael Collins
Associate Chair, Undergraduate Studies
Department of Mechanical and Mechatronics Engineering, University of Waterloo
200 University Ave W.
Waterloo, Ontario. N2L 3G1

Dear Professor Collins,

This report, entitled 'DESIGN OF TEST STAND FOR HIGH RISK ROCKET ENGINE TESTING', was prepared to fulfill my 3A work term requirement. This is my second work term report. I am doing this as a self study on a project I have done for Waterloo Rocketry Design team.

Waterloo Rocketry is a design team that researches, designs, and builds sounding rockets to compete in a yearly competition. As part of the rocket development process, the team performs various static fire tests, where the rocket engine is fired and the performance is measured. The old testing setup was deemed unsafe after propellant research, so a new testing site was found, and a new test setup was needed. My task was to design the new test setup and this report details the design process of this new test setup.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I thank the Waterloo Rocketry Team members for their advice, guidance, and help in manufacturing this.

Sincerely,



Hamza Abuabah

ID: 20710907

3A Mechanical Engineering

Table of Contents

List of Figures	iii
List of Tables	iv
Summary	v
1. Introduction	1
2. Problem background	2
2.1. Safety Assessment and Test site transition	2
2.2. Hybrid engine static fire setup	3
2.3. Replacing the current test stand	4
2.4. Requirements	5
2.1. Design constraints	5
2.1.1. Constraints from containment vessel	5
2.1.2. Constraints from explosion and fire risk	7
2.1.3. Constraints from the weather	7
2.2. Rocketry Test stands	7
2.3. Design alternatives	8
2.4. Test stand modes of failure	9
2.5. Selection of an appropriate test stand design	9
3. Analysis	9
3.1. Structure	10
3.2. Placement and dimensional constraints	11
3.3. Fabrication and assembly	11
3.4. Analysis assumptions	12

3.5.	Analysis.....	14
3.6.	Points of failure.....	15
3.6.1.	Failure by tensile stress	15
3.6.2.	Joint failure	16
3.6.3.	Shipping container failure	19
3.6.4.	Probable modes of failure.....	19
4.	Implementation	19
4.1.	Assembly of the test stand.....	19
4.1.1.	Assembly difficulty	20
4.1.2.	Issue: Lack of corrosion protection	21
4.2.	Shipping container defects	21
4.3.	Explosion.....	22
5.	Conclusions.....	22
6.	Recommendations.....	23
	References.....	24
	Appendix A – Test Stand Main Structure Calculations	27

List of Figures

Figure 1 E5 test setup The ION light rail station can be seen in the background [3]	2
Figure 2 Setup for hybrid engine testing showing all the equipment developed over the years.....	3
Figure 3. Test stand during a static fire of the Kismet engine.....	4
Figure 4 Exploded view of a typical shipping container [4]	6
Figure 5 Labelled isometric drawing of the test stand with all its components.....	9
Figure 6 Test stand steel tube structure with joints and fasteners	10
Figure 7 Model of the shipping container as it would be installed in the shipping container	11
Figure 8 Exploded view of test stand tubes	12
Figure 9 Wireframe diagram of one side of the test stand.....	12
Figure 10 The 3 bodies of the test stand with the forces acting on them	14
Figure 11 Face joints fastening large steel tubes together	16
Figure 12 Cross member to large steel tube joint	17
Figure 13 The reference area used for tear-out calculations.....	18
Figure 14 Photo of Test stand installed into the shipping container	20
Figure 15 Shipping container C-channels in Waterloo Rocketry's shipping container	21
Figure 16 FBD of entire test stand.....	27
Figure 17 FBD of back pillar.....	27
Figure 18 FBD of front pillar	29
Figure 19 Calculation screenshot. Only yellow fields are relevant, the rest is input as described in the report	30
Figure 20 Screenshot from calculations. Only yellow fields are relevant, the rest is input as described in the report.....	31
Figure 21 Screenshot from calculations. Only yellow fields are relevant, the rest is input as described in the report.....	31
Figure 22 Screenshot from calculation of stresses in B-H.....	32
Figure 23 Screenshot from calculation of stresses in G-A	32

List of Tables

Table 1 Requirements for new test stand design.....	5
Table 2 Morphological chart for Engine Support Structure designs [9].....	8

Summary

As part of Waterloo Rocketry's transition to high risk rocket engine testing, the team made the decision to design and fabricate a new rocket engine test stand to replace the previous E5 test stand. This was done to enable the safe testing of liquid engines. New requirements were drafted for the new test setup, and a test stand was designed and built out of structural steel to meet the requirements for an engine support structure for the test setup. The test stand was installed into a shipping container testing vessel in a new engine testing site, and the test stand was used for testing the team's experimental liquid engine. The test stand's design was validated after surviving an accident with the liquid engine with minimal damage to its surroundings.

The test stand discussed will be used for Waterloo Rocketry's future liquid engine tests. Some defects were noted in the shipping container's structure that require the team to do further analysis to validate the safety of testing high power engines on the new test stand. In addition, the shipping container will need to be firmly anchored to the ground to minimise the chance of unintended motion during a static fire test. In addition, the test stand requires a coating of anti-rusting paint to improve its longevity.

1. Introduction

The Waterloo Rocketry Team is a student design team based in the University of Waterloo that researches, develops, and launches sounding rockets to fly in the annual Intercollegiate Rocket Engineering Competition (IREC). The team researches and develops their own rocket engines every year, aiming for larger engines with higher performance. This allows the rockets to fly to higher apogees and allows the team to advance its mission to teach new students crucial engineering skills.

Rocketeers use various types of engines in their rockets. Conventional off-the-shelf model rockets are typically solid rocket motors, which consist of a solid fuel 'grain' containing all the chemicals required for combustion. A solid rocket motor is designed so that, once ignited, it will burn at a specified rate until all the propellant is consumed. This is a simple rocket design with no moving parts, where no plumbing is required. The risks associated with designing and firing a solid motor rocket are low [1, p. 8]. Commercial aerospace rockets typically use liquid propellant rocket engines, which use pressurised liquids to propel a rocket. Most liquid propellant engines are bipropellants, using a liquid fuel and a liquid oxidiser, fed into a combustion chamber by pressure or by pump to achieve a controlled combustion [1, p. 7]. Liquid engines are complex, with various systems required to operate them, so they are much more likely to explode. A hybrid rocket engine is a compromise between the two, as it consists of a solid fuel grain and a liquid oxidiser. It achieves higher specific impulses (thrust/unit mass) than solid motors, with the added complexity of incorporating a liquid feed system.

A crucial part of engine development is rigorous testing to measure the performance of the engine and validate any engineering changes. Waterloo Rocketry typically performs 2 tests for its rocket systems. The first test, referred to as a Cold Flow (CF) test, is where fluids with similar physical properties to the propellants are run through the system in order to diagnose issues, identify leaks, and characterise the flow through the system. Once this is done without issues, the team typically moves on to a Static Fire (SF) test, where the engine is fired for data.

The Waterloo Rocketry team is transitioning from hybrid rocket engines to liquid engines, and this means higher-risk static fire tests. In this transition, a new test setup is needed. The structural part of the test setup that the rocket engine/motor mounts to is typically referred to as a test stand. The objective of this report is to detail the design of the main structure of the new test stand assembly that Waterloo Rocketry will be using to replace the hybrid engine test stand located behind the Engineering 5 (E5) building on the University of Waterloo campus. The scope of this report will be restricted to the main structure of the test stand and will not discuss project management or other parts of the test stand.

2. Problem background

2.1. Safety Assessment and Test site transition

Waterloo Rocketry has typically performed cold flows and static fire tests behind the E5 building, enabling easy access to all the resources of E5's Student Design Centre (SDC). The test setup is shown in Figure 1. As rocket engines increase in size, and new developments appeared in the test site's vicinity, safety concerns were raised about the viability of this test site. The team assessed the danger of testing behind E5 and concluded that it was unsafe to perform SF tests in that location, citing potential property damage to E5, and issues with the proximity to the newly built ION rail [2]. Team operations would need to be relocated to new site away from buildings and human activity. As part of this transition, a new test setup and test stand is needed.



Figure 1 E5 test setup The ION light rail station can be seen in the background [3]

2.2. Hybrid engine static fire setup

Waterloo Rocketry has typically flown hybrid rocket engines. The test setup is made up of a test stand, outfitted with various test equipment developed over the years. Figure 3 shows a hybrid engine (Kismet engine) installed on the E5 test stand. The Combustion Chamber (CC) is where the fuel grain is, and the blast shield serves as a barrier to minimise shrapnel from potential CC explosions. The hybrid CC is installed into a dolly that aligns the engine, and this dolly is bolted to the test stand's structure. Close to the test stand structure is the injector valve, which controls the flow of oxidiser, and it is connected to plumbing, shown in Figure 2.

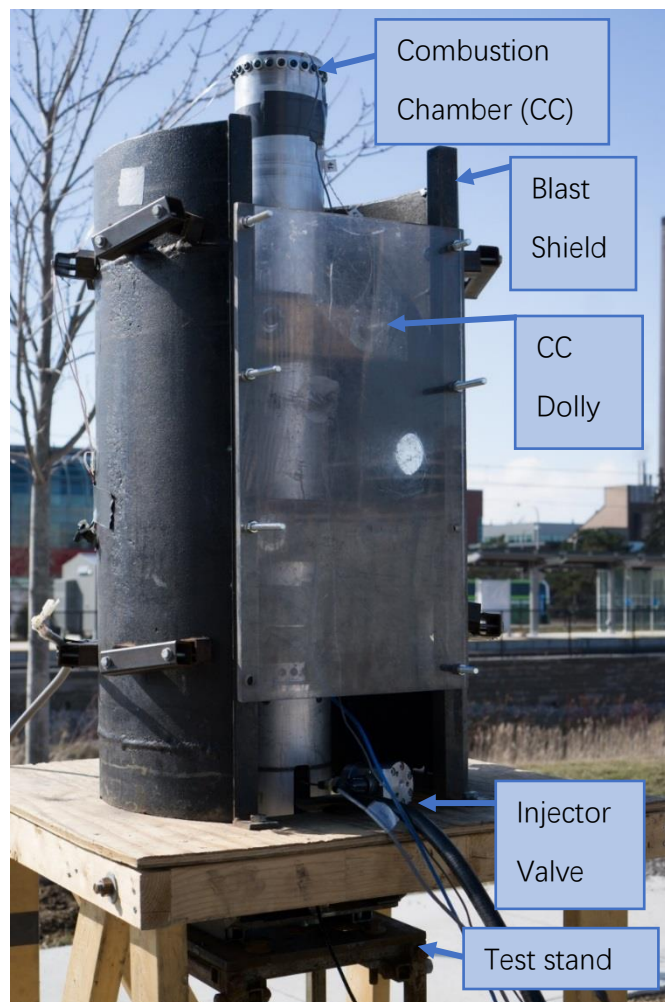


Figure 2 Setup for hybrid engine testing showing all the equipment developed over the years



Figure 3. Test stand during a static fire of the Kismet engine

2.3. Replacing the current test stand

The team can acquire the resources to move the E5 test stand to the new test site, but there are several reasons why this test stand should not be reused for future testing:

- The current test stand is designed for firing a rocket engine upside down (firing upwards). This is not an issue for hybrid engines, but it is an issue for liquid engines. If there are propellant flow timing issues, too much liquid can pool up inside the CC and ignition of this mixture can result in an explosion (typically referred to as a 'hard start'). This is not likely to happen in a hybrid engine because of the simplicity of the feed system in comparison to a liquid engine. To minimise the risk of a liquid engine hard start, the engine can be fired upwards or at an angle of descent, so that propellant leaks out of the CC instead of pooling inside it.

- The current test stand is heavily corroded, and there are some cracks along the length of the tubes of the test stand. As Waterloo Rocketry moves towards engines with higher thrusts, ensuring structural integrity of the test stand is crucial in order to keep SF tests safe.

A test stand will need to be designed to accommodate for the team's current path of hybrid engine development, and account for future liquid engine developments.

2.4. Requirements

The test stand will have new requirements to correct the deficiencies of the previous test stand and expand the testing capacity for future tests. The following requirements were developed:

Table 1 Requirements for new test stand design

No.	Requirement	Relation	Value	Units	Comments
1	Angle of descent of rocket CC	$\geq$	30	deg	This is based on the converging angle of the nozzle, which is 30deg. This would allow excess liquid to leak out and minimise the chance of explosion.
2	Test stand thrust capacity	$\geq$	5000	lbf	Based on recommendation from senior team member.
3	Test stand thrust capacity safety factor	$\geq$	2	N/A	Based on recommendation from senior team member.
4	CC diameter	= (range)	4 - 6	in	CC diameters for liquid and hybrid engines.

2.1. Design constraints

There are several constraints imposed by moving to the new test site and the period in which this test stand is being commissioned. These constraints were considered during the design of the test stand and presented a significant challenge to the implementation of this project.

2.1.1. Constraints from containment vessel

Waterloo Rocketry made the decision to install the new test setup into a shipping container, which will be placed in the new test site. The purpose of the shipping container is to contain shrapnel in the case of any test equipment explosions. The test stand would need to be

installed inside the container. It would be simpler and more secure to install the test stand to the shipping container itself rather than anchor it to the ground, which would involve using ground anchors and require vigorous soil analysis to validate. There is the option of digging into the ground and pouring concrete, but this was not permitted at the new test site. Thus, the most viable option is to mount the test stand to the metallic structure of the container.

The team did not have access to the shipping container for most of the design phase of the test stand, so research had to be done on shipping containers, and the design was to be adaptable to possible variations in shipping containers. The test stand design would only be finalised after the shipping container is obtained to account for possible variations in the shipping container from manufacturing. It was found through research that all shipping containers house C-channels beneath the floor in order to reinforce the wooden flooring of the shipping container [4]. These C-channels were selected as the mounting point for the test stand. The C-channels are shown in Figure 4. The flooring of the container would be removed around where the test stand would mount, and the test stand would be installed into that location by fastening or bonding it to the C-channel cross members.

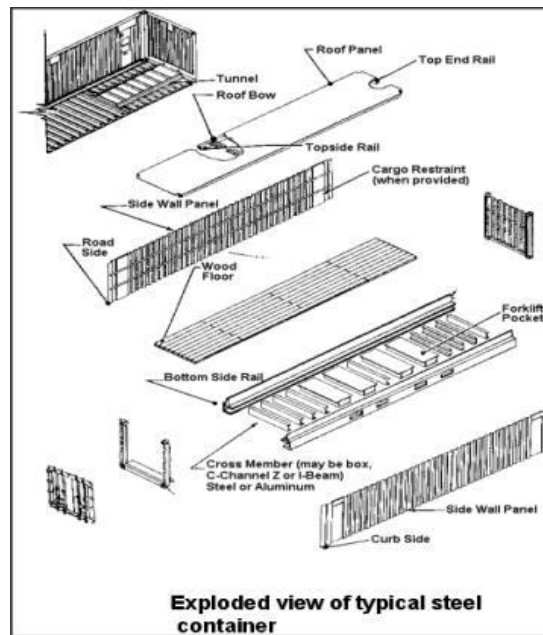


Figure 4 Exploded view of a typical shipping container [4]

2.1.2. Constraints from explosion and fire risk

The test stand needs to be mounted as close to the door of the shipping container as possible. There are many risks associated with having an explosion in a contained vessel. If an explosion is contained, the pressure waves from the explosion can reflect off the walls of the vessel, and the wave intensity may amplify in magnitude and result in an explosion with a higher overpressure [4, pp. sec3-pg2]. This may result in the shipping container walls yielding and/or turning into shrapnel. The rocket engine is the most likely source of an explosion during an SF, so it is beneficial to place it as close as possible to the entrance. This would minimise the amount of pressure waves reflecting off the walls of the shipping container. In addition to mitigating a catastrophic explosion, placing the rocket engine close to the door will protect the flooring of the shipping container. Based on the requirements, the engine will be installed into the test setup at an angle of descent. Its close proximity to the outside will mean that the engine will fire towards the soil, and not the flooring of the container. A barrier (flame diverter) can be added on top of the soil to avoid damaging it.

2.1.3. Constraints from the weather

Test stands tend to use concrete to increase the strength of the stand and allow the steel members to withstand higher stresses, since concrete has a high compressive strength. Unfortunately, the test stand is to be commissioned in February. Historically, the temperature in the Waterloo region is at an average of -2°C [6] in the month of February, which is well below the minimum temperature required to cure cement, which is 10°C for most commercial mixes [7]. This limits the team from using any concrete in the construction of the test stand.

2.2. Rocketry Test stands

Rocketry test setups are an important part of a dedicated rocketry organisation. Such a structure enables the users to perform static fire with their own resources. They typically consist of the following components [6]:

- **Load cell:** The sensor that measures thrust from the engine and sends it to DAQ (Data Acquisition) so that it can be monitored by Rocketry's operations personnel. Data can be later processed to obtain thrust and approximate the performance of the engine
- **Engine Mount:** The actual means by which the engine is secured to the test stand. This mechanism will directly secure the engine to the engine support structure
- **Engine Support Structure:** The structure that absorbs the load of the engine firing and transmits it to a weight or to the anchor points
- **Weights and/or anchor points:** These ensure that the engine stays in place during firing and does not fly away. This can be the ground or large concrete blocks

This report will be strictly about the main test stand structure, the Engine Support Structure.

2.3. Design alternatives

Table 2 Morphological chart for Engine Support Structure designs [9]

Description	Photo
Structural steel test stand, optimised for loading conditions of the test firing	
Heavy metal or salvaged heavy machine structure (Examples: old lathe body or cast iron structure)	
The engine support structure can be the weight. This is usually applicable in the case of a heavy concrete structure or structural steel like large I-beams	

2.4. Test stand modes of failure

The engine support structure, in addition to enabling testing, should be able to account for the rocket engine failing to perform as intended, and avoiding major damage that may result from poorly designed or defective engines. There are several ways a static fire test may fail:

- **Catastrophic explosion** caused by ignition sequencing or propellant flow issues. May cause injury to personnel, damage of test site, or damage of team equipment.
- **Unintended side loading** caused by the engine firing to the side or a hole opening to the side of the combustion chamber, this thrust is a fraction of the nominal thrust of the engine, and likely to cause damage to team equipment/test stand.
- **Escape of the combustion chamber** from the test stand resulting from a poor engine holder design or poor test stand structure. This can be as bad as a catastrophic explosion, depending on the structural integrity of the test stand.

2.5. Selection of an appropriate test stand design

The support structure was selected to be a structural steel design, which is readily available to student design teams, and it is simple to design. All the test stand designs in 2.3 are viable for preventing a failure, so the most convenient and cheapest alternative was selected.

3. Analysis

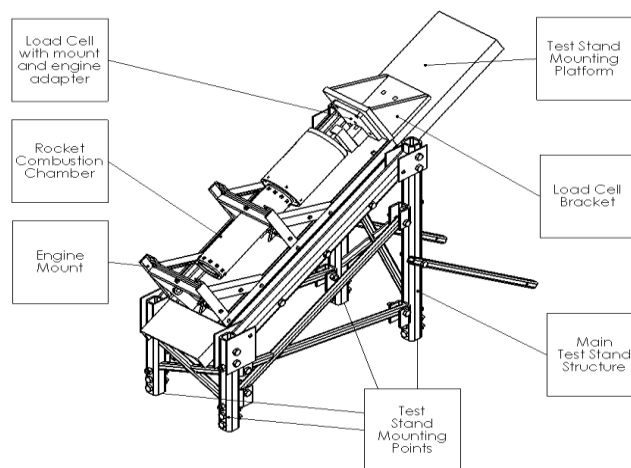


Figure 5 Labelled isometric drawing of the test stand with all its components

The combustion chamber is held in place by an engine mount that is secured onto the dolly salvaged from the E5 test setup. This is secured on a test stand mounting platform, which consists of a 1.5 m long, 10-in wide C-channel that was obtained from the Waterloo Civil Engineering Lab. The C-channel is bolted to the Engine Support Structure, which is made from square steel tubing, bolted to the shipping container. Analysis was done on the Engine Support Structure to optimise it for the required load. A failure of this support structure is defined as causing injury or destruction of non-engine and non-test stand components. This can occur by yielding or dismantlement of the structure, whether the engine on it functions as intended or not.

3.1. Structure

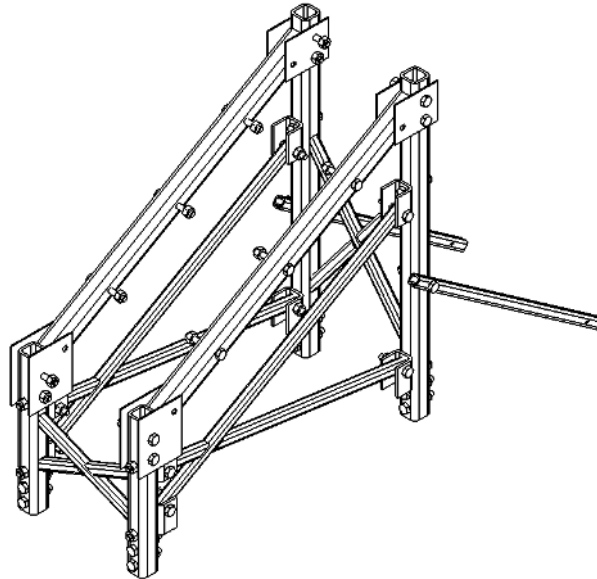


Figure 6 Test stand steel tube structure with joints and fasteners

The main structure of the test stand is set to be made of steel tubes. This allows it to be relatively lightweight while having a higher bending strength than a rod with the same amount of material. Square steel tubes were selected over any other type of structural steel for ease of assembly and welding of steel tubes together, decreasing manufacturing time.

The test stand will be using 4 large 2"x2"x1/4" square steel tube pillars, and small 1"x1"x1/8" square steel tube cross members between the pillars. There are additional steel tube reinforcements extending from the back pillars of the test stand to the ground to reduce the

bending moment in the longer back steel pillars. This structure was designed to sustain a load of 5000 lbf with a Factor of Safety (FOS) of 2, a design load of 10,000 lbf for the test stand structure.

3.2. Placement and dimensional constraints

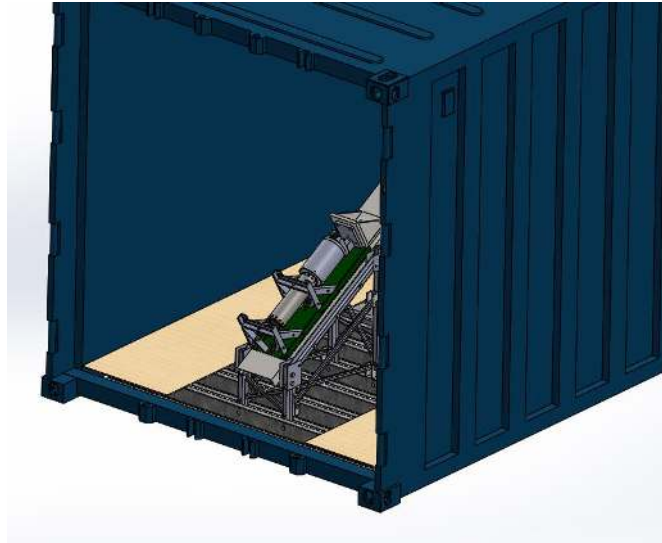


Figure 7 Model of the shipping container as it would be installed in the shipping container

The exact dimensions of the structure will depend on its placement in the container. The height will depend on the distance from the end of the shipping container to the front of the test stand. The height of the back of the test stand will be dictated by the height of the front pillars and the angle that the test stand is set at. It is adjusted based on a CAD model of the test setup that includes a shipping container.

The test stand would be placed towards the entrance of the shipping container to minimise the amount of shrapnel created and prevent the shipping container from melting or being set on fire. The engine fires slightly outside of the container, where some sort of flame diverter can be installed. The dimensions were adjusted based on assembly considerations, ensuring that it can be bolted into the C-channels of the container.

3.3. Fabrication and assembly

The structure of the test stand will be split up into weldments, and then assembled on site into the shipping container. The weldments were split so that it would be possible to modify the cross

members according to the shipping container obtained. It is also done so that there is some adjustability in the setup. Adjusting the angle of descent of the engine would involve fabricating new cross members and rails, then disassembling the test stand and reassembling it into a new configuration.

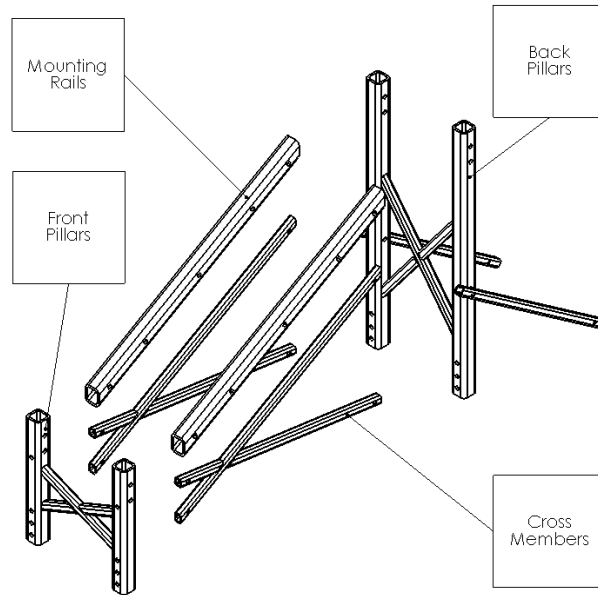


Figure 8 Exploded view of test stand tubes

3.4. Analysis assumptions

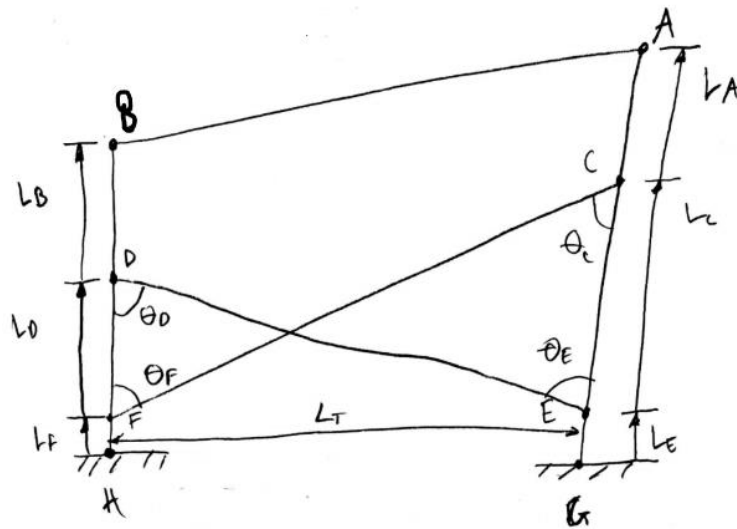


Figure 9 Wireframe diagram of one side of the test stand

Half of the test stand will be analysed, with one half of the design load, i.e. 5,000 lbf. To analyse the test stand structure, some assumptions will need to be made to simplify the analysis and make the problem solvable. The following assumptions were made, with reasoning:

- The load on the test stand is evenly distributed across both sides of the test stand. This is a reasonable assumption since the rocket engine is centered on the test stand and fires straight.
- The force will be assumed to act along beam A-B, which is a reasonable assumption given that the engine will be firing parallel to that line. The beam A-B will be bolted to the Test Stand Mounting Platform. If the engine fires straight and there are no forces going perpendicular to A-B, this is a valid assumption.
- For the purposes of simplifying analysis, beam A-B will not be considered to exist. The force acting on beam A-B will be split up between point A and point B. This is not an accurate assumption, since the beam A-B will resist deformation and strengthen the structure. This is a conservative assumption, which is fine for the purposes of this analysis.
- The test stand will be considered as having 3 bodies. These are B-H, A-G, and beam CDEF.
- The points C, D, E, F are all pin joints, so there is no moment applied at those joints
- The point G is a fixed point and the point H is a pin joint. This will approximate conditions since beam A-G will be more constrained than beam B-H, since there are support beams attached to A-G. This is an approximation to simplify analysis. It will result in a smaller moment in beam B-H.. This is a conservative assumption because A-G will fail before B-H.

The rocket load will be split between point A and point B, but the exact distribution is uncertain. It is immediately apparent that the rocket will be firing closer to point A towards point A, which means that a larger portion of the load is acting on point A. If there is 2500 lbf acting on B-A, then the majority will act on point A. The load at point A is at least 50% of the thrust. However, it is uncertain how much bias there is towards the front. The test stand will be designed for between 50-60% of bias at point A. The steel tube reinforcements are added in part to strengthen beam A-G.

To solve the test stand problem, 3 equations were made for the 3 different bodies, B-H, A-G, and C-D-E-F. With these, the forces at each point were determined. These forces were incorporated into a beam analysis to determine the points of highest stress in the beam. The highest stresses were computed, and the test stand dimensions were optimised for the design load of 10,000 lbf and assumption that the load at point A is 50-60% of the thrust. The full calculation can be found in Appendix A.

3.5. Analysis

The test stand structure was separated into 3 bodies:

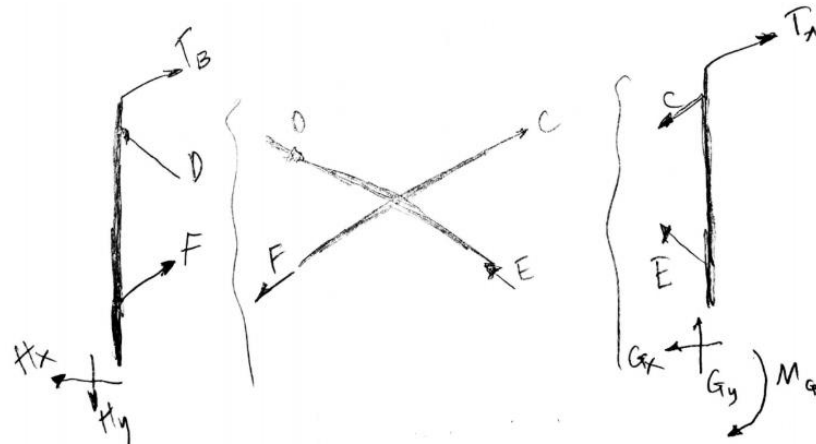


Figure 10 The 3 bodies of the test stand with the forces acting on them

Each body was analysed, and the maximum shear and axial stresses were computed. The maximum force at any of the pin joints was determined to be 4,900lbf at joints C and D. The dimensions of C-D-E-F were modified until the structure was optimised for the 10,000lbf design load (5,000lbf for one side), which would allow for the 5,000lbf thrust with a FOS of 2.

The body C-D-E-F, if viewed as a block, is under a shear loading. The points/beams C and F are under tensile stress, and D and E are under a compressive load. Cross members are optimal for this kind of loading since they resist shear loading.

3.6. Points of failure

3.6.1. Failure by tensile stress

The test stand can fail by tensile loading at beam A-G (A-G will fail before B-H because it is longer and under a higher load). The bending moment from the rocket firing towards the beam and the tensile stress from the rocket firing upwards combine to create a large axial stress on the beam. To calculate the axial stress, the combined loading equation was used (equation 1). The moment M is the moment in the beam, and the force F is the tensile force on the beam. y is the max distance from the neutral axis of the beam to the face of the beam (1/2 the height of the beam), and A and I are the cross-sectional area and the moment of inertia of the cross section, respectively.

$$\sigma = \frac{F}{A} + \frac{My}{I} \quad (1)$$

This will compute the maximum stress on the beam, which must stay below the yield stress of the structural steel. The steel used in the beams is A36 steel, which has a yield stress of 36,000psi. [9]

The beams in C-D-E-F are also under an axial stress, but since they are pin jointed- the moment is assumed to be 0. The axial load on these beams was found to be 4,900lbf using equation 1, and this is well below the yield stress of these members, which is 15,750 lbf for a compressive load for tensile stress, and 9,625 lbf for a 1"x1" square steel tube with a thickness of 1/8". The cross members are in danger of buckling as well.

$$F = \frac{\pi^2 EI}{(kL)^2} \quad (2)$$

The maximum load for buckling can be calculated using equation 2. E is the modulus of elasticity of the material, I is the moment of inertia of the tube, L is the length of the beam, and k is a factor accounting for end conditions [10]. The loading condition for this scenario is a beam that is fixed at one end (the welded end) and pin jointed at the other (to the pillar), which would mean a k -factor of 0.7. However, if the joint is loose, this can result in a less

optimal condition, fixed on one end and rounded/loose at the other end. In this scenario, the k-factor is 2, which is worst case scenario. This value was used for the calculation. The value used for L was the length of the longest beam in the cross members, which is 17 inches long. The E value used is 29,000ksi, which is typical for A36 steels [9]. The I for a 1"x1" square steel tube with a thickness of 1/8" was used. This is approximately 0.05697in^4 . The critical buckling load is 14,100lbf for worst case conditions, so these beams are unlikely to buckle during loading. Given this analysis, the test stand is highly unlikely to fail by axial loading of tubes.

3.6.2. Joint failure

The test stand has multiple types of joints. There are face joints attaching the 2"x2" square steel tubes together. These face joints consist of plates that have holes in them

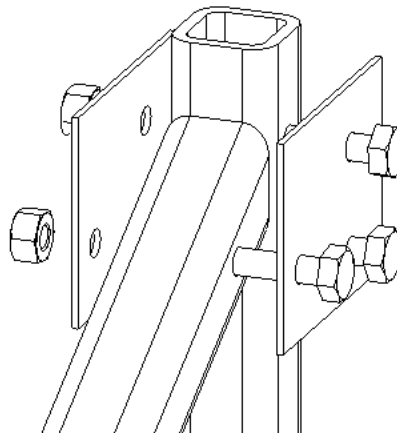


Figure 11 Face joints fastening large steel tubes together

These joints are relatively nonstructural, since they only carry the load of the engine's mass. Most of the load is concentrated on the pillars. The large steel tubes are in contact with each other, so forces transfer by direct contact rather than through the face joint plates.

The other type of joint in the test stand are the joints between the cross members and the large steel tubes. These joints consist of a modified square steel tube, with a bolt attaching the joint to the large steel pillar, and another bolt acting as a pin to hold a small tube from the cross member.

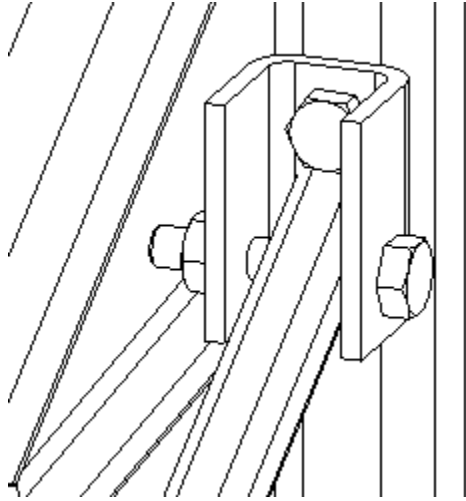


Figure 12 Cross member to large steel tube joint

This joint is structural and likely to fail if designed improperly. This joint is present at points C, D, E and F on the test stand. This means that the maximum load on this is 4,900 lbf. The load-stress relation is:

$$\sigma = F/A \quad (3)$$

The joint is likely to fail in the following ways:

- **Tear-out at the small steel tube**

Tear-out can happen when the tension on the small steel tube in the joint is large enough so that an entire chunk of steel is sheared. This is computed using equation 3, where σ is the shear yield stress of the material, F is the tensile force on the steel tube. A is the shear area. For the analysis, the minimum distance from the edge was computed given the maximum tensile force of 4900 lbf with this equation, and the shear yield stress of A36 steel, which is approximately 20,880 psi (yield stress * 0.58) [9]. The minimum edge distance to avoid tear out was computed to be 0.71in, so the joints were designed with that edge distance.

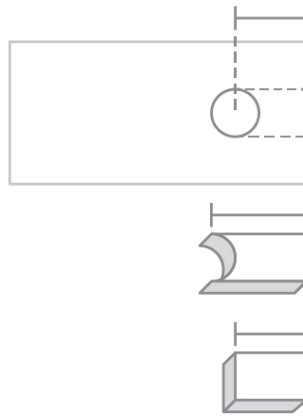


Figure 13 The reference area used for tear-out calculations

- **Bearing failure at the cross-member steel tube**

This mode of failure happens when there is a force on the edge of the hole from the bolt. This force creates a stress on the edges of the hole, and this can result in localised buckling of the tube, a bearing failure. The maximum force for bearing failure can be computed using equation 3, where σ is the yield stress of the material, F is the force on the steel tube. A is the bearing area, which is the area under stress between the bolt and the steel tube. The maximum force given the conditions was computed to be 4,320 lbf, which is below the maximum force the joints will experience at maximum loading. This means that the joints are at risk of failing by a bearing failure. Fortunately, this is not a catastrophic mode of failure, and since the test stand is not undergoing fatigue loading, the failure of the test stand from this joint bearing failure is unlikely. To minimise the chance of this happening, the joints must be inspected after every SF test to ensure that there is no deformation.

- **Shearing of the bolt**

The fasteners used are $\frac{1}{2}$ " bolts Grade 5 bolts. The minimum yield strength of this bolt is 92,000 psi. The maximum shear force on the bolt in double shear can be computer with $\sigma = F/A$, where σ is the shear yield stress of the bolt, which is approximately 48,760psi (yield stress * 0.58) [11]. A is the bolt's shear reference area, which is the bolt's cross-sectional area multiplied by 2, since it is in double shear. This results in a maximum force of

14,361 lbf, which is well above the maximum force on the joint, so it is unlikely to fail by shearing of the bolt.

3.6.3. Shipping container failure

The test stand is bolted to the shipping container, which is assumed to be structurally enough to withstand the loading by the SF. This must be put through a separate analysis to verify and is not within the scope of this report.

3.6.4. Probable modes of failure

- The structure fails by the tube walls being damaged by a localised bearing failure. This will not result in a catastrophic failure mode
- The shipping container's C-channels fail by a tensile stress or shear tear out of the test stand from the C-channels
- The test stand steel tubes fail by tensile/compressive yielding caused by a large moment or load on the tubes. This is unlikely to occur for engine sizes being tested

4. Implementation

4.1. Assembly of the test stand

In February 2020, the Waterloo Rocketry team obtained a shipping container and modified it to include a steel flooring and added various equipment to it so that the shipping container can be used for a liquid engine test at the end of February. Towards the end of the modifications period, the test stand was manufactured and then installed into the shipping container.



Figure 14 Photo of Test stand installed into the shipping container

4.1.1. Assembly difficulty

The bolt holes were designed with a tight tolerance for a $\frac{1}{2}$ " bolt, to maximise the contact area between the bolt and the tube. This was to reduce the chance of bearing failure. This was done with minimal consideration for assembly difficulty and tolerance stack-ups during the welding process. It was extremely difficult to assemble the test stand, and some parts of it were not possible to assemble without using a hammer to force bolts into holes. This resulted in a large amount of stripped nut and bolt threads, making it difficult to disassemble in case there is an issue with the stand, or a part needs to be replaced. This is a defect with the test stand. It can be addressed by using a drill with a larger size to ream out all the holes to a larger size for a $\frac{1}{2}$ " clearance size hole.

The installation of the test stand into the container was difficult because of the spacing between the shipping container C-channels. To install the test stand into the shipping container, a drill needed to be found that would fit between the shipping container C-channels, and a drill bit short enough to create the holes. A step drill was used in order to

drill into the shipping container C-channels since It minimises retooling time and it is sharp enough to drill quickly into the shipping container C channels.

4.1.2. Issue: Lack of corrosion protection

One of the issues with the hybrid engine test stand was the corroded condition of the test stand, which caused cracks to form along the length of the steel tubes. Adding anti-corrosion coating will increase the service life of this equipment and prevent the growth of moss and other flora inside the steel tubes of the test stand, which is directly exposed to grass [8].

4.2. Shipping container defects



Figure 15 Shipping container C-channels in Waterloo Rocketry's shipping container

It was noted that the shipping container's C-channels were corroded in some spots, which may compromise its properties. This is fine for testing smaller engines, but the issue must be explored in detail before testing larger engines. The first static fire test done with the new test stand was with a 600 lbf engine, which is a small fraction of the force the test stand was designed for.

In addition, the shipping container's C-channels were warped and the spacing between them was inconsistent, which made assembly difficult.

4.3. Explosion

The test stand was used for the first time at the end of Feb 2020 for a 600 lbf liquid engine static fire test. Unfortunately, due to propellant pressurisation issues, the engine exploded. This resulted in major damage to the CC and all parts connected to it. The test stand survived relatively unscathed. The engine mount was slightly damaged, with the only failure being the shearing of a small bolt. This is relatively minor equipment damage, so the test stand is considered a success.

5. Conclusions

The following conclusions can be drawn from the analysis and execution of this project:

1. The E5 test stand is no longer safe for testing, so the new test stand should be used in its place for testing of high risk rocket assemblies: The success of the test stand in the liquid engine test, passing through the test with minor damages, demonstrates that this test stand is suitable to use for Waterloo Rocketry's future testing.
2. The testing vessel obtained for use in the test setup has mounting points that are not ideal for usage in the test setup. This means that the analysis of this test stand is flawed, and that while the test stand structure itself may be strong enough to withstand the loading required, the shipping container test vessel may be unsuitable for higher engine thrusts.
3. The shipping container is currently not ready for the higher power engines, since it is not fixed to the ground. The shipping container is currently placed on the ground without

any anchoring, which means it can only rely on the force of friction to prevent it from moving.

4. The test stand is not ready for outdoor conditions, and it is susceptible to corroding like the old test stand.

6. Recommendations

There are several recommendations for Waterloo Rocketry, moving into the future with this new test stand:

1. The usage of the shipping container C-channels must be re-evaluated. The C-channels can be reinforced, or the mounting points can be changed.
2. The shipping container must be anchored to the ground for higher power engine tests.
3. The new test stand must be coated with a non-combustible, anti-rust paint that will protect it from the corrosion of the outdoors.

References

- [1] G. P. Sutton, Rocket Propulsion Elements, Hoboken, New Jersey: John Wiley & Sons Inc., 2017.
- [2] A. Leszkowiat, "Engine Testing and Propellant Safety Analysis," Waterloo Rocketry Team, Waterloo, ON, 2019.
- [3] Waterloo Rocketry, "November 11 2018 Engine Test (SF3)," 11 November 2018. [Online]. Available: <https://www.flickr.com/photos/uwrocketry/45122962184/in/album-72157673410944607/>.
- [4] J. Pike, "FM 55-17 Chapter 15 Container Construction and Inspection," globalsecurity.org, 2020. [Online]. Available: <https://www.globalsecurity.org/military/library/policy/army/fm/55-17/ch15.htm>. [Accessed 10 May 2020].
- [5] Department of Homeland Security, "FEMA-426 / BIPS-06: Reference Manual to Mitigate Potential Terrorist Attacks Against Buildings," October 2011. [Online]. Available: <https://www.dhs.gov/xlibrary/assets/st/st-bips-06.pdf>. [Accessed 10 May 2020].
- [6] AccuWeather, "Waterloo, Ontario, Canada Monthly Weather - February," AccuWeather, Inc, 2020. [Online]. Available: <https://www.accuweather.com/en/ca/waterloo/n2j/february-weather/55073>. [Accessed 14 May 2020].
- [7] Portland Cement Association, "Cold Weather Concreting," 2019. [Online]. Available: <https://www.cement.org/learn/concrete->

technology/concrete-construction/cold-weather-concreting. [Accessed 10 May 2020].

- [8] R. Nakka, "Richard Nakka's Experimental Rocketry Site: Rocket Motor Static Testing," Nakka Rocketry, 25 July 2018. [Online]. Available: <https://www.nakka-rocketry.net/static.html>. [Accessed 10 May 2020].
- [9] Aerocon, "Aerocon Test Stand Pictorial," Aerocon Systems, 2020. [Online]. Available: http://www.aeroconsystems.com/ts_pics/ts_pics.htm. [Accessed 17th May 2020].
- [10] Metal Supermarkets, "Grade Guide: A36 Steel," 14 November 2017. [Online]. Available: <https://www.metalsupermarkets.com/grade-guide-a36-steel/>. [Accessed 16 May 2020].
- [11] Engineering Tool Box, "Euler's Buckling Formula," 2012. [Online]. Available: https://www.engineeringtoolbox.com/euler-column-formula-d_1813.html. [Accessed 17 May 2020].
- [12] Bolt Depot, "Bolt Markings and Strength Chart," Bolt Depot, 2020. [Online]. Available: <https://www.boltdepot.com/fastener-information/materials-and-grades/bolt-grade-chart.aspx>. [Accessed 17 May 2020].
- [13] Coating.ca, "Anti Corrosion Coating - Anti Rust Coating," Coating.ca, 2020. [Online]. Available: <https://coating.ca/anti-corrosion-coating/>. [Accessed 14 May 2020].
- [14] E. Mohrman, "What Is the Altitude of a Plane in Flight?," USA TODAY, 28 March 2018. [Online]. Available: <https://traveltips.usatoday.com/altitude-plane-flight-100359.html>. [Accessed 9 May 2020].

- [15] N. Christopher, T. Cojocar, J. Deery, R. Jiang, D. Kletke, R. Kobets, S. Kong, S. Martin, Z. Mears, A. Mihaila, A. Morrison, V. Rajkumar, J. Robinson, K. Tam, T. Tang and E. Yang, "Shark of the Sky Hybrid Rocket: Team 45 Project Technical Report for the 2019 IREC," in *Proceedings of the 2019 Spaceport America Cup Conference*, Las Cruces, NM, 2019.

Appendix A – Test Stand Main Structure Calculations

Feb 17th 2019

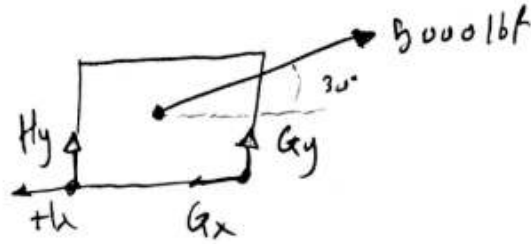


Figure 16 FBD of entire test stand

$$\Sigma F_y = 0$$

$$0 = 5000 \sin 30^\circ + H_y + G_y$$

$$-2500 = H_y + G_y \quad (1)$$

$$\Sigma F_x = 0$$

$$0 = 5000 \cos 30^\circ - H_x - G_x$$

$$4330 - G_x - H_x = 0 \quad (1)$$

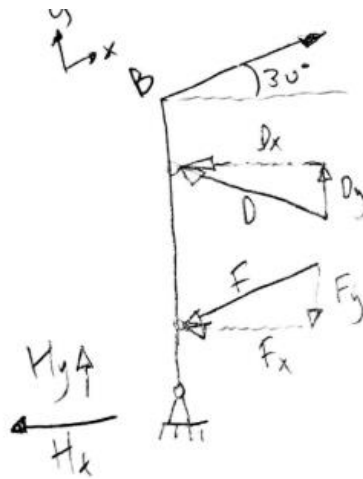


Figure 17 FBD of back pillar

$$\Sigma F_x = 0$$

$$0 = T_B \cos 30^\circ - D_x - F_x - H_x \quad (3)$$

$$\sum F_y = 0$$

$$0 = T_B \cdot \sin 30^\circ + D_y - F_y + H_y \quad (4)$$

$$\sum M_H = 0$$

$$0 = F_x \cdot L_F + D_x \cdot L_D - T_B \cdot \cos 30^\circ \cdot L_e \quad (5)$$

Assume the value of **H_x**. It is 40-50% of the 4330lbf according to (1) and assumptions stated earlier in the report. Use equations (3) and (5) to find **D_x** and **F_x**. (2 equations and 2 unknowns)

Assumption, D and F are single force members

$$\tan \theta_F = \frac{F_x}{F_y}$$

$$F_y = \frac{F_x}{\tan \theta_F} \quad (6)$$

$$\tan \theta_D = \frac{D_x}{D_y}$$

$$D_y = \frac{D_x}{\tan \theta_D} \quad (7)$$

Find **F_y** and **D_y** using equations (6) and (7). Plug them into (4) and (1) to find **H_y** and **G_y**.

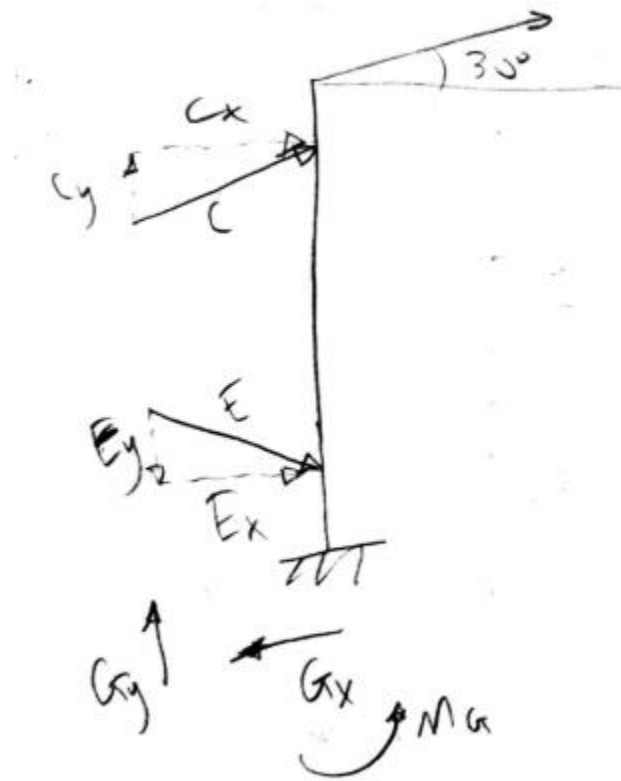


Figure 18 FBD of front pillar

$$\Sigma F_x = 0$$

$$0 = T_A \cos 30^\circ + C_x + E_x - G_x \quad (8)$$

$$\Sigma F_y = 0$$

$$0 = T_A \sin 30^\circ + C_y - E_y + G_y \quad (9)$$

$$\Sigma M_G = 0$$

$$0 = M_G - E_x \cdot L_E - C_x \cdot L_C - T_A \cdot \cos 30^\circ \cdot L_A \quad (10)$$

Use (8) and (9) to find **Cx** and **Ex**. Use 10 to find **Mg**. All variables should now be known.

FINDING Hy USING EQUATION 4					
Equation	Fy	Dy	Tbsin30		
4	-580.92	2449.24	1250		
results					
Hy	-4280.15 lbf				
FINDING Gy USING EQUATION 2					
Equation	Hy	(T)sin30			
2	-4280.15	2500			
results					
Gy	1780.15 lbf				
FINDING Cy AND Ey USING EQUATIONS 8 AND 9 WITH MATRICES					
Equation	Cy coeff	Ey coeff	coeff		
8	1.74963712	7.376765505	0		0.10957
9	1	-1	-3030.15258		0.10957
results					
Cy	-2449.24 lbf				
Ey	580.92 lbf				
FINDING Cx AND Ex USING SINGLE FORCE MEMBER ASSUMPTION					
Equation	Force	Angle			
11	-2449.24	60.25			
12	580.92	82.28			
results					
Cx	-4285.28 lbf		C magnitude	4935.823789 lbf	
Ex	4285.28 lbf		E magnitude	4324.471777 lbf	
FINDING Mg using equation 10					
Equation	Ex*Le	Cx*Lc	Ta*La*cos30		
10	9213.344179	-82691.34966	62440.43161		
Mg	-11037.57387 lbf*in				

Figure 19 Calculation screenshot. Only yellow fields are relevant, the rest is input as described in the report

BEAM B-H SHEAR (USING MACAULY FUNCTIONS)			
Equation	Hx	Fx	Dx
Comparison	2165.06	4285.28	4285.28
Hx	2165.06		Shear stress max
Hx-Fx	-2120.2		6582.37 lbf
Hx-Fx+Dx	2165.06		
BEAM B-H MAX MOMENT (USING MACAULY FUNCTIONS)			
Hx*Lf	2489.82		
Hx*Ld-Fx*(Ld-Lf)	-8768.5		
Hx*Lb-Fx*(Lb-Lf)+Dx*(Lb-Ld)	0		
Axial Stress from Moment			
	11769.80834		psi
Axial Stress from Hy			
	2834.538132		psi
Combines axial stress			
	14604.34647		psi
Failure is at 36,000 so the tube is good			

Figure 22 Screenshot from calculation of stresses in B-H

BEAM G-A SHEAR (USING MACAULY FUNCTIONS)				
Equation	Gx	Dx	Cx	Mg
Comparison	2165.06	4285.28	4285.28	-11038
Gx	2165.06			Shear stress max
Gx-Dx	-2120.21			6446.02 lbf
Gx-Dx+Cx	2165.06			
BEAM G-AMAX MOMENT (USING MACAULY FUNCTIONS)				
-Mg	11037.6			
-Mg+Gx(Ld)	25023.9			
-Mg+Gx(Lc)-Dx(Lc-Ld)	-2192.5			
	18469.5			
Axial Stress from Moment				
	33589.10623			psi
Axial Stress from Gy				
	1780.15			psi
Combines axial stress				
	35369.26			
Failure is at 36,000 so the tube is good				

Figure 23 Screenshot from calculation of stresses in G-A